

Skills Summary

Normal-Model z-Scores and Percentiles

Use this reference while completing your worksheet or when studying.

Skill 1 — Standardising: from a Value to z

Formula: $z = \frac{x - \mu}{\sigma}$ — how many standard deviations the value sits above ($z > 0$) or below ($z < 0$) its own mean.

Order matters: it is value minus mean, never mean minus value. Getting it backwards flips the sign and turns a below-average result into an above-average one.

Why bother: z strips the units off, so two results measured on completely different scales can be compared.

Example: A test has $\mu = 70$ and $\sigma = 5$. Standardise a score of 78.

Step 1. The question asks how unusual one value is inside its own distribution, which is exactly what standardising measures, so use the z formula.

Step 2. $z = \frac{78 - 70}{5} = \frac{8}{5}$.

$\Rightarrow z = 1.6$, so the score is 1.6 standard deviations above the mean.

Try it: Same test ($\mu = 70$, $\sigma = 5$). Standardise a score of 62.

check: $z = -1.6$

Skill 2 — From z to a Percentile

Every entry below is the area to the LEFT of that z — the proportion of the distribution at or below it. Multiply by 100 and you have the percentile. For an area to the RIGHT, subtract the entry from 1.

z	-2.00	-1.50	-1.28	-1.00	-0.50	0.00
area left	0.0228	0.0668	0.1003	0.1587	0.3085	0.5000
z	0.50	0.67	1.00	1.28	1.50	2.00
area left	0.6915	0.7486	0.8413	0.8997	0.9332	0.9772

Example: What proportion lies below $z = 0.50$, and what proportion lies above it?

Step 1. The table only ever reports left-tail area, so read the left answer straight off and get the right-hand one by complement.

Step 2. Left: the entry at 0.50. Right: 1 minus that entry.

below = **0.6915**, above = **0.3085**

Try it: What proportion lies below $z = -0.50$, and what proportion lies above it?

check: below = 0.3085, above = 0.6915

Skill 3 — From a Percentile Back to a Value

Reverse formula: $x = \mu + z\sigma$. Multiply by σ FIRST — adding z straight onto μ is the most common mistake in this direction.

Method: (1) turn the percentile into a left-tail area; (2) find the z whose table entry is closest to it; (3) substitute into $x = \mu + z\sigma$.

Top $p\%$? That is a right-tail statement. Convert it first: area to the left = $1 - p/100$.

Example: For $\mu = 70$, $\sigma = 5$, which value sits at the 75th percentile?

Step 1. A percentile names an area, not a value, so find the z with that area to its left and then undo the standardising.

Step 2. Closest table area to 0.75 is **0.7486** at $z = 0.67$; then $x = 70 + 0.67(5) = 70 + 3.35$.

$\Rightarrow x = \mathbf{73.35}$

Try it: For $\mu = 70$, $\sigma = 5$, which value sits at the 10th percentile?

check: area to the left of $z = -1.28$ is 0.10

Skill 4 — Comparing Results Across Distributions

Never compare raw values from different distributions. Standardise each one against its OWN mean and its OWN standard deviation, then compare the two z -scores: the larger z is the relatively stronger result.

Say it in context: “ $z = 1.5$ means that result is 1.5 standard deviations above the mean of its own group.”

Example: 78 on a test with $\mu = 70$, $\sigma = 5$, or 88 on one with $\mu = 80$, $\sigma = 8$?

Step 1. The two scales are different, so the raw numbers cannot be compared; standardise each result inside its own distribution instead.

Step 2. First: $z = 1.6$ from Skill 1. Second: $z = \frac{88 - 80}{8}$.

second $z = \mathbf{1.0}$, and $\mathbf{1.6} > \mathbf{1.0}$, so the first result is stronger.

Try it: Is 66 on a test with $\mu = 60$, $\sigma = 4$ stronger than a result with $z = 1.2$?

check: $z = 1.5$ and $1.2 < 1.5$, so yes